

Advanced Statistical Methods II

Lecture 11: From Kaplan–Meier to Regression for Survival Data

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Outline of Today's Lecture

From one survival curve to many covariates

Choosing what to model

The proportional hazards idea

The Cox proportional hazards model

Hazard ratio interpretation

Risk sets and the intuition behind Cox estimation

Examples and model interpretation

When the Cox model is reasonable

Summary and next steps

- Observed survival data: (Y_i, δ_i) , where

$$Y_i = \min(T_i, C_i), \quad \delta_i = \mathbb{I}(T_i \leq C_i).$$

- ▶ Risk set at event time t_j : individuals still under observation and event-free just before t_j .
- ▶ Kaplan–Meier estimator:

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

A motivating medical example



Suppose a cancer study records time to relapse.

Patient	Time	Event?	Treatment	Age	Stage
1	5.2	1	New	42	II
2	8.1	0	Control	67	III
3	3.4	1	New	71	III
4	10.7	0	Control	50	II
⋮	⋮	⋮	⋮	⋮	⋮

If we draw Kaplan–Meier curves for treatment groups, are we answering the whole question?

Not necessarily. Treatment groups may differ in age, stage, biomarker levels, or baseline prognosis.

Why Kaplan–Meier is not enough



Kaplan–Meier is excellent for estimating **one survival curve** or comparing a few groups.

But it becomes limited when many covariates are present.

- ▶ It handles groups, but not continuous covariates naturally.
- ▶ Splitting age into age groups loses information and creates arbitrary cut-points.
- ▶ With several covariates, stratified KM curves create many small cells.
- ▶ It does not give a compact covariate effect such as “per 10-year increase in age.”

The curse of many Kaplan–Meier curves



Suppose we stratify by three covariates:

Covariate	Categories	Number of strata
Treatment	New / Control	2
Age group	Young / Middle / Old	3
Disease stage	I / II / III / IV	4
Total		$2 \times 3 \times 4 = 24$

Groups versus regression



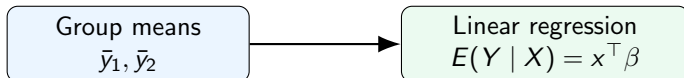
Group comparison	Regression modelling
“Do the survival curves differ between groups?” Usually categorical groups	“How does survival change with covariates?” Categorical and continuous covariates
KM curves, log-rank test	Cox model, parametric regression models
Mostly unadjusted	Can adjust for multiple covariates
Visual comparison	Estimates, standard errors, confidence intervals

Regression does not replace Kaplan Meier. It answers a different, more covariate-focused question.

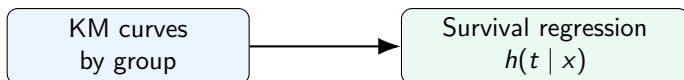
A familiar analogy



For ordinary continuous responses:



For survival responses:



What is the survival-analysis analogue of the regression function?

To model the **hazard function**.

Three possible targets



For survival data, a covariate may affect several related quantities.

Target	Symbol	Interpretation
Survival function	$S(t x)$	Probability of surviving beyond t
Density	$f(t x)$	Distribution of event times
Hazard function	$h(t x)$	Instantaneous event rate at t

The hazard is

$$h(t | x) = \lim_{\Delta t \downarrow 0} \frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t, x)}{\Delta t}.$$

It is a natural target because it describes the event risk among individuals who are still event-free.

Why model the hazard?



The hazard is local in time.

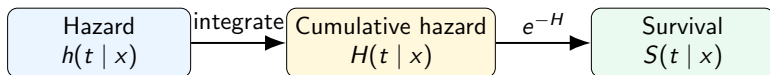
- ▶ At time t , it focuses on individuals who have survived up to t .
- ▶ It allows covariates to act on the instantaneous event rate.
- ▶ It provides a useful effect measure: the **hazard ratio**.

The Cox model keeps the risk-set idea and adds a regression structure for covariates.

then

$$H(t \mid x) = \int_0^t h(u \mid x) du,$$

$$S(t \mid x) = \exp\{-H(t \mid x)\}.$$

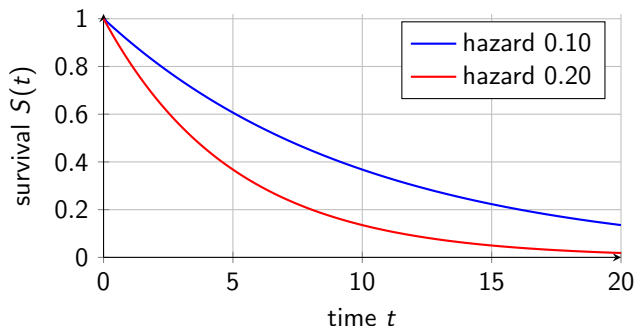


So a model for $h(t \mid x)$ automatically implies a model for $S(t \mid x)$.

A small visual intuition



Suppose a baseline group has constant hazard $h_0(t) = 0.10$. A higher-risk group has hazard 0.20.



A larger hazard means a faster decline in survival, but the survival curves themselves are not separated by a constant vertical distance.

The proportional hazards idea



For two covariate values x_a and x_b ,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)} \text{ does not depend on } t.$$

The shapes over time are controlled by a common time pattern; covariates multiply that pattern.

Multiplicative rather than additive effects



A proportional hazards model is multiplicative:

$$h(t | x) = h_0(t) \times \text{relative risk depending on } x.$$

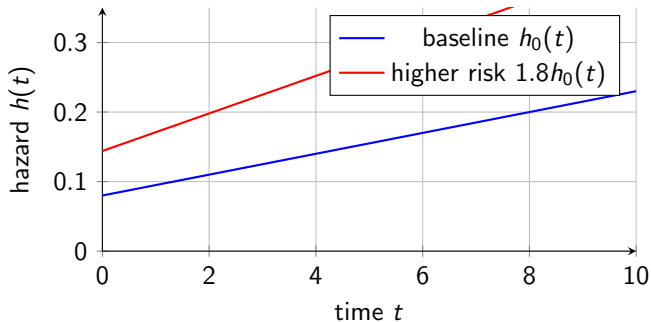
For example, if treatment reduces hazard by 30%, then

$$h(t | \text{new treatment}) = 0.70 h(t | \text{control})$$

for all t , under proportional hazards.

This does **not** mean survival probabilities are 30% larger at all times.
The hazard ratio is a ratio of instantaneous event rates.

Visualizing proportional hazards



The two hazard curves can change over time, but their ratio is always 1.8.

Question	Model component
How does baseline risk evolve over time?	$h_0(t)$
How do covariates change risk?	$\exp(x^\top \beta)$

This is why the Cox model is often called **semiparametric**.

where

- ▶ $h_0(t)$ is an unspecified baseline hazard;
- ▶ $\mathbf{x} = (x_1, \dots, x_p)^\top$ is a vector of covariates;
- ▶ $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$ is the regression coefficient vector.

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What does baseline hazard mean?



If $x = 0$, then

$$h(t \mid x = 0) = h_0(t).$$

So $h_0(t)$ is the hazard for the baseline covariate profile.

Interpretation

If x contains treatment ($1 = \text{new}, 0 = \text{control}$) and age centered at 60, then $h_0(t)$ is the hazard for a control patient aged 60.

If covariates are not centered, $x = 0$ may not be scientifically meaningful. The hazard ratios are usually more important than the literal baseline profile.

Why the exponential form?



The model uses

$$\exp(x^T \beta)$$

for the relative risk.

This has several advantages:

- ▶ It is always positive, so hazards remain nonnegative.
- ▶ Regression effects become additive on the log-hazard scale:

$$\log h(t | x) = \log h_0(t) + x^T \beta.$$

- ▶ Coefficients have hazard-ratio interpretations.

Linear predictors are convenient; exponentiation converts them into multiplicative hazard effects.

A model with two covariates



Suppose

$$x_1 = \mathbb{I}(\text{new treatment}), \quad x_2 = \text{age in decades above 60}.$$

Then

$$h(t \mid x_1, x_2) = h_0(t) \exp(\beta_1 x_1 + \beta_2 x_2).$$

Coefficient	Meaning	Hazard ratio
β_1	New treatment versus control	e^{β_1}
β_2	10-year increase in age	e^{β_2}

Example

If $\beta_1 = -0.357$, then $e^{\beta_1} = 0.70$: the new treatment has 30% lower hazard than control, adjusting for age.

Cox model implies proportional hazards



For two individuals with covariates x_a and x_b ,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)} = \frac{h_0(t) \exp(x_a^\top \beta)}{h_0(t) \exp(x_b^\top \beta)}.$$

Therefore,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)} = \exp\{(x_a - x_b)^\top \beta\}$$

The baseline hazard cancels. The hazard ratio does not depend on time.

Connection to survival curves



Let

$$H_0(t) = \int_0^t h_0(u) du, \quad S_0(t) = \exp\{-H_0(t)\}.$$

Under the Cox model,

$$H(t | x) = H_0(t) \exp(x^\top \beta),$$

so

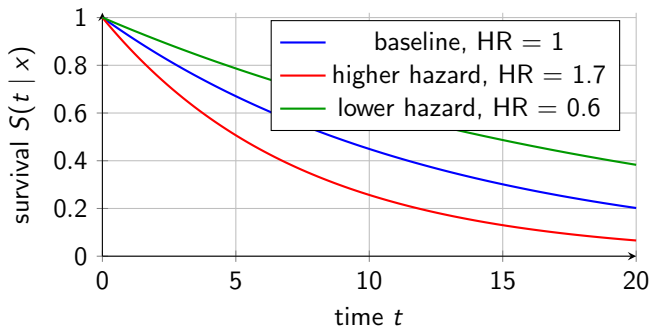
$$S(t | x) = S_0(t)^{\exp(x^\top \beta)}.$$

The model creates a family of survival curves by transforming one baseline survival curve.

Survival curves under a Cox model



Here is a stylized example with $S_0(t) = \exp(-0.08t)$.



The survival curves are not parallel; the proportionality is on the hazard scale.

$$\text{HR}_k = e^{\beta_k}.$$

A hazard ratio is not a probability ratio, not an odds ratio, and not directly a ratio of median survival times.

Example: treatment effect

Suppose the Cox model is

$$h(t \mid x) = h_0(t) \exp(\beta x),$$

where $x = 1$ for new treatment and $x = 0$ for control.

If $\hat{\beta} = -0.5108$, then

$$e^{\hat{\beta}} = e^{-0.5108} \approx 0.60.$$

Interpretation

At any time t , among patients still event-free just before t , the new-treatment group has about 40% lower instantaneous relapse rate than the control group.

This interpretation assumes proportional hazards.

Example: age effect



Suppose age is measured in years and

$$\hat{\beta}_{\text{age}} = 0.035.$$

Then the one-year hazard ratio is

$$\exp(0.035) = 1.036.$$

A one-year interpretation is often too small to communicate clearly. For a 10-year increase,

$$\exp(10 \times 0.035) = \exp(0.35) = 1.42.$$

A 10-year older patient has about 42% higher instantaneous event rate, holding other covariates fixed.

Example: continuous biomarker



Let x be a biomarker standardized to mean 0 and variance 1.

If

$$\hat{\beta} = 0.50,$$

then

$$\exp(0.50) = 1.65.$$

Interpretation

A one standard deviation increase in the biomarker is associated with a 65% higher hazard, adjusting for other covariates.

For continuous covariates, always say what a “one-unit” increase means.

Comparing two covariate profiles



For two patients a and b ,

$$\text{HR}(a : b) = \exp\{(x_a - x_b)^\top \beta\}.$$

Suppose

$$\beta_{\text{treat}} = -0.36, \quad \beta_{\text{age10}} = 0.30.$$

Compare:

- ▶ Patient A: new treatment, age 70;
- ▶ Patient B: control, age 60.

Then

$$\log \text{HR}(A : B) = (-0.36)(1) + (0.30)(1) = -0.06,$$

so

$$\text{HR}(A : B) = e^{-0.06} = 0.94.$$

Treatment benefit and age disadvantage nearly cancel.

A common misinterpretation



Suppose $HR = 0.60$ for treatment.

Wrong statement: “The treated group has 40% higher probability of survival at every time.”

Better statement:

At each time t , among individuals who are still at risk just before t , the treated group has 40% lower instantaneous event rate than the control group, under the proportional hazards model.

The survival-probability difference depends on time and on the baseline hazard.

The Cox model also uses risk sets.

$$\mathcal{R}(t_j) = \{i : Y_i \geq t_j\}.$$

The Cox model asks: among the individuals in the risk set, whose covariates make them more likely to be the one who fails now?

A small risk-set example



At time $t = 6$ months, suppose patient 2 fails. The risk set is:

Patient	In risk set?	Treatment	Linear score $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$
1	Yes	Control	0.00
2	Yes	New	-0.40
3	Yes	Control	0.30
4	No, censored earlier	New	—
5	Yes	New	-0.10

The Cox model assigns relative risk score

$$\exp(\eta_i).$$

The failed subject is compared only with subjects in the same risk set.

Why the baseline hazard cancels



At an event time t_j , suppose exactly one individual fails.

Under the Cox model,

$$h_i(t_j) = h_0(t_j) \exp(\mathbf{x}_i^\top \beta).$$

Given that someone in $\mathcal{R}(t_j)$ fails at t_j , the approximate probability that subject i is the one who fails is

$$\frac{h_0(t_j) \exp(\mathbf{x}_i^\top \beta)}{\sum_{\ell \in \mathcal{R}(t_j)} h_0(t_j) \exp(\mathbf{x}_\ell^\top \beta)}.$$

Thus

$$\boxed{\frac{\exp(\mathbf{x}_i^\top \beta)}{\sum_{\ell \in \mathcal{R}(t_j)} \exp(\mathbf{x}_\ell^\top \beta)}}$$

The unknown $h_0(t_j)$ cancels.

Example 1: clinical trial



Outcome: time to disease progression.

Covariate	Possible coding
Treatment	1 = new drug, 0 = standard care
Age	years, or decades above 60
Stage	indicator variables for stages III and IV
Biomarker	standardized value

A possible model:

$$h(t | x) = h_0(t) \exp(\beta_1 \text{Trt} + \beta_2 \text{Age}_{10} + \beta_3 \text{Stage}_{\text{III}} + \beta_4 \text{Stage}_{\text{IV}}).$$

The treatment hazard ratio is adjusted for age and disease stage.

Example 2: reliability engineering



Outcome: time until machine failure.

Possible covariates:

- ▶ operating temperature;
- ▶ vibration level;
- ▶ load level;
- ▶ manufacturer;
- ▶ maintenance schedule.

A Cox model might be

$$h(t | x) = h_0(t) \exp(\beta_1 \text{Temp} + \beta_2 \text{Vibration} + \beta_3 \text{Load}).$$

Interpretation

If $e^{\beta_1} = 1.25$ per 10 degrees, then each 10-degree increase in temperature is associated with a 25% higher failure hazard, holding vibration and load fixed.

Example 3: customer churn



Outcome: time until a customer leaves a service.

Possible covariates:

- ▶ monthly price;
- ▶ number of support tickets;
- ▶ usage frequency;
- ▶ plan type;
- ▶ region.

Survival methods are not only for medicine. They apply whenever the response is a time until an event, with censoring.

Here, censoring may mean the customer was still active when the dataset was closed.

Covariate	Estimate	Hazard ratio	Interpretation?
Treatment	−0.45	0.64	?
Age10	0.28	1.32	?
BiomarkerSD	0.52	1.68	?

Write one sentence interpreting each hazard ratio. Be precise about “holding other covariates fixed.”

- Do not say “36% lower probability of death at every time.” The model is about hazard, not direct survival probability.

Assumption	Meaning
Proportional hazards	Covariate hazard ratios are constant over time
Log-linear covariate effect	$x^\top \beta$ is appropriate on the log-hazard scale
Independent censoring	Censoring is not informative after conditioning on covariates

The Cox model avoids specifying $h_0(t)$, but it still imposes structure on covariate effects.

What proportional hazards does not mean



Proportional hazards does not mean:

- ▶ survival curves are parallel;
- ▶ survival differences are constant over time;
- ▶ median survival times have a fixed ratio;
- ▶ the treatment effect is causal without a suitable study design or adjustment strategy.

It means that for two fixed covariate profiles,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)}$$

is constant in t .

When proportional hazards may fail



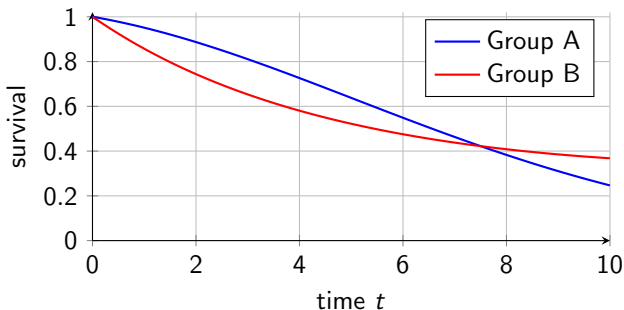
Examples where proportional hazards may be questionable:

- ▶ A surgery has high early risk but long-term benefit.
- ▶ A drug works strongly at first but its effect weakens later.
- ▶ Two KM curves cross substantially.
- ▶ A machine component has burn-in failures and then stabilizes.

If the hazard ratio changes over time, a single Cox hazard ratio may be an oversimplification.

Possible remedies include time-varying coefficients, stratified Cox models, or alternative survival models.

A visual warning: crossing survival curves



Crossing or converging curves suggest that the relative hazard may not be constant over time.

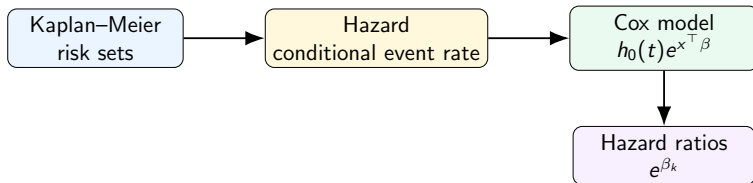
How KM still helps after Cox modelling



Kaplan–Meier remains useful even when regression is needed.

- ▶ Initial exploratory plots by major groups.
- ▶ Visual check for crossing curves.
- ▶ Communication of unadjusted survival experience.
- ▶ Comparison with model-based adjusted survival curves.

A good survival analysis often uses both KM curves and regression models.



The Cox model is best understood as risk-set-based regression for event times.

Key Takeaway Messages



1. KM curves estimate survival nonparametrically, but they do not easily adjust for multiple covariates.
2. Survival regression models the event-time distribution conditional on covariates.
3. The Cox model specifies

$$h(t | x) = h_0(t) \exp(x^\top \beta),$$

with unspecified baseline hazard.

4. Hazard ratios compare instantaneous event rates among those still at risk.
5. The proportional hazards assumption means hazard ratios are constant over time.
6. Risk sets provide the bridge from Kaplan–Meier estimation to Cox partial likelihood.

What comes next



Topic	Why it matters
Partial likelihood	Estimating β without specifying $h_0(t)$
Ties	Multiple events at the same observed time
Wald / score / likelihood ratio tests	Inference for covariate effects
Schoenfeld residuals	Checking proportional hazards
Adjusted survival curves	Communicating fitted models

Today: model intuition. Next: estimation and diagnostics.

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